

Designing for Rapid Decisions: Eye-Tracking and Usability Analysis of an Emergency Department Sepsis Dashboard

Peter Wagner*

FM&H, SOVS, SPRL
UNSW

UNSW - Sydney, Australia
p.wagner@unsw.edu.au

Shailendra Sawleshwarkar

Sydney Medical School
University of Sydney

Sydney, NSW, Australia
shailendra.sawleshwarkar@sydney.edu.au

Soojeong Yoo*

Health and Immersive Interaction (HII) Lab
University of Sydney

Sydney, NSW, Australia
soojeong.yoo@sydney.edu.au

Audrey P Wang

Biomedical Informatics and Digital Health
University of Sydney

Westmead, New South Wales, Australia
audrey.wang1@sydney.edu.au

Abstract

We present an in-situ usability evaluation of the SAFE-WAIT Risk Identification Dashboard (SWRID), an AI-powered clinical decision support system (CDSS) used by triage nurses to assess sepsis risk in Emergency Department waiting rooms. Nine nurses completed clinically realistic tasks while participating in Think-Aloud protocols, Single Ease Question ratings, and unobtrusive remote eye-tracking. This mixed-methods approach captured both conscious decision-making strategies and unconscious attention patterns in a high-pressure environment. Findings showed reliance on a small set of high-salience indicators, with gaze data confirming these priorities and revealing task-specific differences in scan paths. This study demonstrates the feasibility of combining gaze and verbal protocol analysis in live clinical settings, translating into actionable design refinements and offering a transferable HCI framework for optimising safety-critical decision support tools where speed, accuracy, and cognitive load management are essential.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *User interface design*; **Empirical studies in HCI**; • **Applied computing** → *Health care information systems*.

Keywords

Clinical Decision Support Systems, Emergency Department, Sepsis, Usability Evaluation, Think-Aloud Protocol, Eye-Tracking, Human-Computer Interaction, Interface Design

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*Both authors contributed equally to this research.



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1 Introduction

Health Information Systems and AI-powered Clinical Decision Support Systems (CDSS) are vital for enhancing patient safety, yet ensuring they minimise cognitive load in high-pressure Emergency Departments (EDs) remains a major HCI challenge [2, 6, 10, 11]. Interfaces that obscure critical information or require unnecessary steps can increase error risk and hinder adoption.

Prior research in HCI has established that 'visual search' is a primary determinant of cognitive load in medical interfaces [7]. However, traditional usability heuristics often focus on static layout rules rather than dynamic workflow integration. By examining 'scan paths'—the temporal sequence of fixations—we can uncover friction points that standard heuristic evaluations might miss, such as the cognitive cost of cross-referencing spatially separated data points [5, 15].

While usability evaluations often utilise Think-Aloud protocols to capture subjective reasoning, these methods may fail to identify unconscious interaction behaviours. Eye-tracking offers a complementary, objective measure of attention and workload [13, 15]. However, most CDSS usability studies occur in controlled simulations [7], limiting their ecological validity. In-situ evaluations under real-world environmental pressures remain rare.

To address this gap, we conducted an in-situ mixed-methods evaluation of an AI-powered sepsis risk dashboard (SWRID) in a live ED [12]. By integrating remote eye-tracking with Think-Aloud protocols, we captured both articulated clinical strategies and unconscious visual patterns. This study demonstrates a transferable framework for evaluating safety-critical systems, ensuring design aligns with the realities of clinical practice.

2 Study Design

To understand how clinical decision support tools influence attention in high-pressure environments, we conducted an *in-situ* mixed-methods evaluation of the SAFE-WAIT Risk Information Dashboard (SWRID). The study took place at a single metropolitan Western Sydney tertiary care hospital involving Acute Care Triage

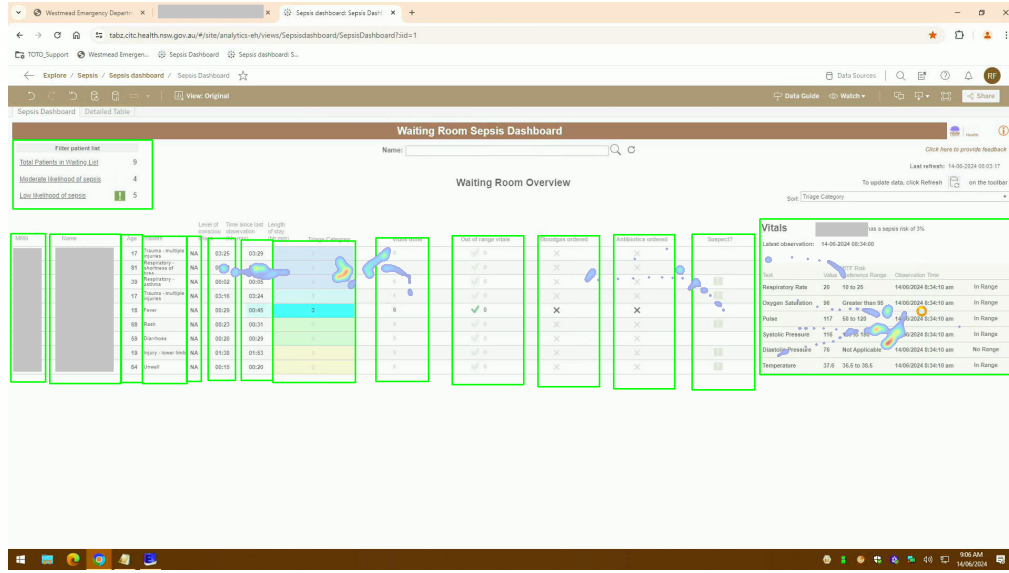


Figure 1: The SWRID interface. To map attention in a live environment, we used OCR to dynamically identify Regions of Interest (ROIs), such as the 'Suspect?' column, overlaying aggregate eye-tracking heatmaps.

nurses responsible for monitoring the Emergency Department waiting room.

We employed three concurrent methods to capture both conscious strategies and unconscious behavioural patterns: a Think-Aloud protocol, the Single Ease Question (SEQ), and remote eye-tracking. Recruitment utilised convenience sampling, engaging nine Acute Care Triage nurses (7 female, 2 male) rostered on morning shifts. Participants possessed varying levels of digital literacy and clinical experience (Range: 2–15 years), ensuring the findings reflect a representative cross-section of the workforce rather than just 'expert' users.

Apparatus & Environment. The SWRID is an AI-powered CDSS integrated into the hospital's Electronic Medical Record (EMR) [4, 12]. It visualises sepsis risk via a colour-coded interface (Figure 1). To maintain ecological validity, we installed a remote GP3-HD eye-tracker (GazePoint, Canada) directly at the active Clinical Workstation. The setup was designed to be unobtrusive; screen content was mirrored to a dedicated recording laptop to ensure data security and prevent interference with hospital IT operations.

Procedure: Clinical Tasks. Participants performed three randomised tasks designed to reflect authentic triage workflows. We explicitly defined these as:

- **Task A (Triage Review):** Identifying patients at risk of sepsis, but not assigned to an appropriate triage category.
- **Task B (Risk Assessment):** Identifying patients with elevated risk scores who require immediate further testing (e.g., blood gas, lactate).
- **Task C (Handover Evaluation):** Assessing the global state of the waiting room to prioritise care, simulating a start-of-shift handover.

Participants verbalised their thought processes (concurrent Think-Aloud) during these tasks and rated perceived difficulty (SEQ) immediately after each one[3].

Data Processing. Qualitative transcripts were analysed thematically to identify decision-making strategies. To account for the dynamic updates inherent to live clinical dashboards, we employed an Optical Character Recognition (OCR) method to automatically track Regions of Interest (ROIs). This approach ensured that fixation data remained spatially accurate even as patient rows shifted during triage workflows.

3 Results

3.1 Data Integrity & Usability Ratings (SEQ)

Of the nine recruited nurses, seven provided complete datasets (Think-Aloud, SEQ, and eye-tracking). Calibration accuracy was robust (mean error = 19.8 pixels), with no significant performance difference between seated and standing postures ($p = .85$).

Subjectively, participants found the system manageable. Median SEQ ratings were high for Task A (Triage Review) and Task C (Handover) ($Mdn = 6$), and slightly lower for the moderate-risk Task B ($Mdn = 5$). A Friedman test confirmed no statistically significant difference in perceived difficulty across tasks ($\chi^2(2) = 3.88, p = .144$). However, qualitative and gaze data revealed distinct underlying strategies.

3.2 Qualitative & Gaze Strategies

Visual Anchoring on Risk Indicators. Across all tasks, the *Suspect?* column served as the primary cognitive anchor. In Task A (identifying mismatch), nurses explicitly prioritised this column alongside *Out of Range Vitals*. One nurse noted, “I would go straight

to the percentages... she has one that's not between the flags." Objective gaze data confirmed this reliance: the *Suspect?* ROI received the largest share of fixation duration, acting as the entry point for deeper investigation.

Complexity in Moderate-Risk Assessment (Task B). While Tasks A and C elicited linear scanning strategies, Task B (identifying moderate risk) prompted more varied behaviour. Participants described a need to "cross-reference" multiple data points. As one explained: "green means low likelihood... exclamation mark means moderate," requiring them to check *Antibiotics Ordered* and *Blood Gas* status. Eye-tracking metrics aligned with this reported complexity. Fixation attention was significantly more distributed in Task B than in A or C, with increased dwell time on secondary diagnostic fields (*Vitals Done*, *Antibiotics*) rather than a singular focus on the risk score.

Scan Paths and Interaction Costs. Transition matrix analysis (Figure 2) revealed that participants predominantly followed sequential scan paths, evidenced by the strong diagonal density (reading rows left-to-right). However, significant off-diagonal clusters indicate high-frequency non-sequential gaze jumps, specifically between *Out of Range Vitals* and *Suspect?*. This pattern suggests a "search-compare-verify" loop where users actively bridge the spatial gap between risk scores and physiological data. A recurring friction point was the dashboard's limited interactivity. Participants frequently attempted to hover or click rows to reveal granular risk data, expressing frustration ("it would be heaps better if it just popped up... instead of having to hover") when the interface did not respond, highlighting a mismatch between user expectation and system feedback.

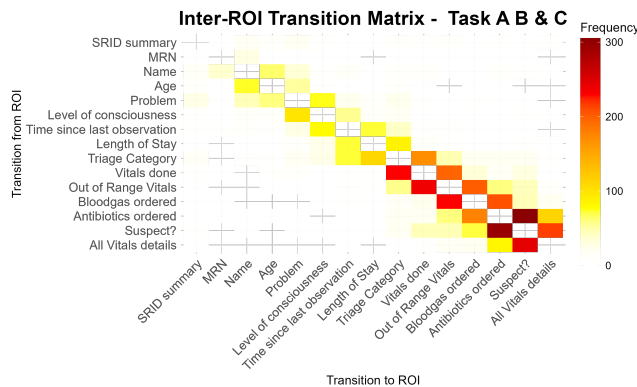


Figure 2: Combined transition matrix of Tasks A, B, and C showing gaze transitions between ROIs. Darker colours indicate more frequent transitions, with a strong diagonal reflecting sequential scanning behaviour across adjacent ROIs.

4 Discussion

This study evaluated the usability of the SWRID in a real-world Emergency Department. By triangulating gaze data with verbal protocols, we identified specific friction points where the system design clashed with clinical reasoning.

4.1 Cognitive Anchoring vs. Distributed Search

Our eye-tracking data confirms that the *Suspect?* column acts as a "cognitive anchor." In high-pressure scenarios (Tasks A and C), nurses rely on this single high-salience cue to rapidly filter patients, a strategy consistent with heuristics used in time-constrained clinical settings [11]. However, this heuristic breaks down during moderate-risk assessments (Task B), where the decision requires synthesising physiological data (Vitals) with algorithmic output (Risk Score). The increased fixation distribution and non-sequential gaze jumps observed in Task B indicate a "Split Attention Effect," where users are forced to hold information in working memory while visually traversing the screen to cross-reference separated fields [6].

4.2 The Cost of Interaction and Hidden Data

The recurring frustration with "hover-to-reveal" mechanics highlights a critical HCI principle: high-frequency safety checks should have zero interaction cost. In a triage context, the millisecond delays caused by hovering to see a risk percentage accumulate, disrupting the scan path and potentially increasing cognitive processing demands during time-critical tasks [6]. The disconnect between the subjective SEQ ratings (which were high) and the objective gaze complexity (which showed friction) reinforces the danger of relying solely on self-report methods in healthcare evaluations; users may report a task as "easy" simply because they are accustomed to working around poor design [9].

4.3 Design Implications for AI-Driven Dashboards

Our findings offer design guidelines applicable beyond sepsis tools to any safety-critical CDSS that pairs an "AI Score" with "Clinical Evidence." The following design implications are derived directly from empirically observed gaze-transition patterns and participant-reported friction points.

The "Evidence-Score" Proximity Rule. A recurring issue was the "Split Attention Effect" observed in Task B [6]. Users were forced to hold the AI's risk prediction (the Score) in working memory while scanning distantly for the physiological data (the Evidence) that justified it. *Implication:* CDSS interfaces should spatially couple algorithmic outputs with their primary determinants. For example, if an AI flags 'High Risk,' the contributing vital signs should be rendered spatially adjacent to the score. This layout mimics the cognitive process of validation, reducing the saccadic travel required to verify the algorithmic output.

Information Radiators vs. Interactive Queries. The frustration with "hover-to-reveal" interactions [6] suggests that in high-pressure environments, dashboards must function as "Information Radiators" displays that convey status without user intervention. *Implication:* Interaction costs, such as clicks or hovers, should be eliminated for primary safety checks. "Drill-down" interactions are appropriate only for secondary, confirmatory data (e.g., historical trends) that do not determine the immediate triage category.

4.4 Methodological Contribution

This work validates the feasibility of *in-situ* multimodal evaluation. Traditional eye-tracking ROI analysis assumes static interface layouts. However, live clinical dashboards involve continuous row shifts as patient data updates, rendering fixed-coordinate ROI mapping unreliable. Our OCR-based dynamic ROI detection enables temporally stable fixation mapping under real-time layout changes. This extends gaze-based HCI evaluation beyond controlled laboratory settings into dynamic, safety-critical environments [14].

4.5 Limitations

Our study was conducted at a single site with a small sample ($N = 9$, 7 complete datasets from nurses), which is common for *in-situ* clinical studies but limits generalisability to other roles such as physicians. The "Think-Aloud" protocol may have artificially slowed performance or influenced gaze behaviour, though it provided necessary context for the gaze data [8]. We did not directly measure diagnostic accuracy, time-to-decision, or clinical error rates. This exploratory study focuses on attentional mechanisms and interaction patterns that precede clinical outcomes. Future research should explicitly link gaze-transition metrics and scan path entropy to measurable performance and safety indicators. Finally, our analysis was descriptive; future work will employ probabilistic modelling (e.g., Hidden Markov Models) to predict scan paths and link specific gaze sequences to clinical error rates [1].

5 Conclusion

We presented an *in-situ* evaluation of an AI-powered sepsis dashboard, revealing how spatial layout influences clinical attention. Findings show that while nurses utilise sequential scanning for general overview, moderate-risk assessments trigger cognitively demanding non-sequential eye movements due to the spatial separation of related data. By shifting from subjective feedback to objective gaze metrics, we identified specific design flaws such as hidden risk percentages and separated vital signs that add friction to decision making. These insights offer an evidence-based roadmap for optimising CDSS interfaces, ensuring they support, rather than hinder, the rapid cognitive work required in Emergency Departments.

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